Math 1200 Lecture 2

Sep 8 2026

Topics: 12.4-12.5: Cross product, equations of lines and planes Name:_____
Collaborators:_____

Group work 1 Find the vector and parametric equations for the line passing through $(4, 2, 1)$ and parallel to the line $x = -t + 3,\ y = 8 - 2t,\ \ z = 2t + 10$.

Group work 2 Which of the following data determines a unique plane?

(a) Two distinct points on the plane

(b) Three distinct points on the plane

(c) A point on the plane and a vector on the plane

(d) A point on the plane and a vector orthogonal to the plane

Group work 3 Find the vector and scalar equations for the plane passing through $P(1,0,1), Q(-3,1,1)$ and $R(2,4,0)$.

Group work 4 Let M be the plane with equation $x-6y+2z=14$. Calculate the shortest distance from the point $Q(3,-2,4)$ to the plane M as follows:

- Find any point P on M. Find the vector $\vec{v}=\vec{PQ}$.

- Argue that the shortest distance from P to M is the magnitude of the projection $\text{comp}_{\vec{n}}\,\vec{v}$ and calculate this distance.